

Vitamin B6 — The Essential Guide

Vitamin B6 (pyridoxine) is a cofactor in over 140 enzymatic reactions, more than almost any other single micronutrient. It's essential to neurotransmitter synthesis, red blood cell formation, and amino acid metabolism — yet mild insufficiency is common and frequently goes unrecognized because early symptoms are vague.

WHAT B6 DOES

— Neurotransmitter synthesis

Required cofactor for producing serotonin, dopamine, GABA, and norepinephrine from their amino acid precursors.

— Red blood cell formation

Involved in hemoglobin synthesis; deficiency can produce a form of anemia distinct from iron or B12 deficiency.

— Homocysteine regulation

Works alongside B12 and folate to metabolize homocysteine, an amino acid linked to cardiovascular risk at elevated levels.

— Immune function

Supports lymphocyte proliferation and interleukin-2 production; deficiency measurably impairs immune response in controlled studies.

SIGNS OF LOW B6

- Fatigue, irritability, or low mood without an obvious cause
- Cracked skin at the corners of the mouth (angular cheilitis) or a swollen, sore tongue
- Numbness or tingling in the hands and feet (peripheral neuropathy) at more severe deficiency
- Confusion or difficulty concentrating

Food Sources & Dosing

FOOD SOURCES (PER TYPICAL SERVING)

Chickpeas, 1 cup cooked	1.1 mg
Salmon, 3 oz cooked	0.6 mg
Chicken breast, 3 oz cooked	0.5 mg
Potato, 1 medium with skin	0.4 mg
Banana, 1 medium	0.4 mg
Fortified cereal, 1 serving	0.5–2.0 mg

RECOMMENDED INTAKE

Adults: 1.3 mg/day (ages 19–50), rising to 1.5–1.7 mg/day after 50. Pregnant: 1.9 mg/day. Tolerable Upper Intake Level: 100 mg/day for adults — doses meaningfully below this are considered low-risk, but B6 is one of the few water-soluble vitamins where high, chronic supplemental doses (commonly above 200 mg/day for extended periods) have been linked in case reports to sensory neuropathy.

SUPPLEMENT FORMS

— Pyridoxine HCl

The standard, inexpensive synthetic form found in most multivitamins. Converted to the active form (P5P) by the liver.

— Pyridoxal-5-Phosphate (P5P)

The already-active coenzyme form. Costs more; theoretically bypasses a conversion step, though clinical significance for most healthy adults is minor.

Who Should Pay Attention

HIGHER-RISK GROUPS

- People taking isoniazid, hydralazine, or certain other medications that interfere with B6 metabolism
- Adults with alcohol use disorder — alcohol accelerates B6 breakdown
- People with kidney disease, malabsorption conditions (celiac, Crohn's), or autoimmune disorders
- Those using oral contraceptives long-term, which is associated with modestly lower B6 status in some studies

SELF-CHECK

- ☐ I eat at least one B6-rich food (poultry, fish, chickpeas, potatoes, bananas, fortified cereal) most days
- ☐ I don't rely on high-dose B6 supplements (over 100 mg/day) without a specific reason
- ☐ I've noted whether any medications I take are known to interact with B6
- ☐ If I supplement, I've checked whether my multivitamin's B6 dose plus food intake could exceed the upper limit
- ☐ I've had bloodwork or discussed B6 status with a clinician if I have persistent, unexplained fatigue or nerve symptoms

This guide is educational and not a substitute for individualized medical advice. Consult a healthcare provider before starting or changing any supplement regimen, particularly at doses above standard multivitamin levels.